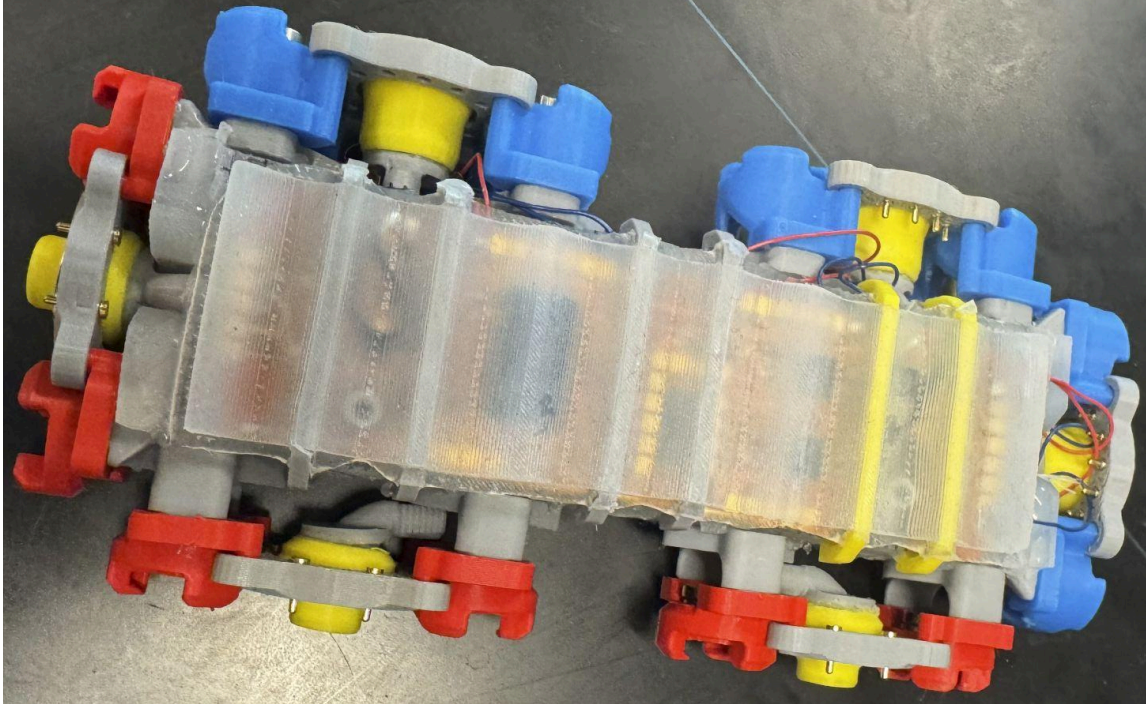


Modular flexible PCB Design for Self-Reconfigurable Soft Robotics

W.E Chase Quijano - Rowan University

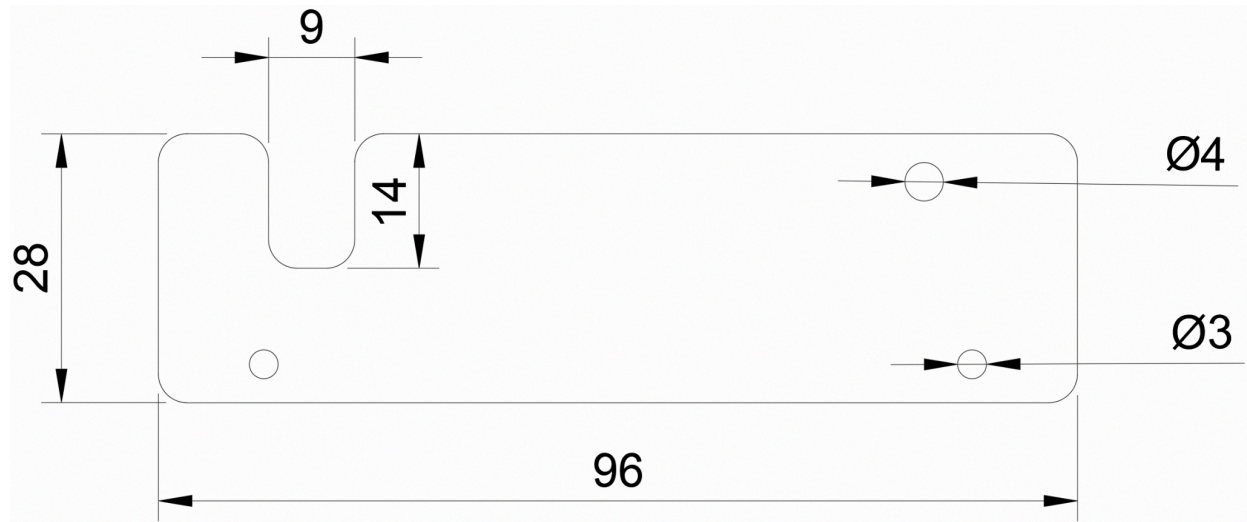


Problem Statement:

Self-reconfigurable soft robotics modules require compact, reliable, and modular electronics to drive actuators, interface with sensors, and enable communication between units.

The challenge: design PCBs that integrate seamlessly with soft robotic hardware while meeting strict space and reliability constraints, including the ability to maintain $\sim 1^\circ$ of flexion during operation.

Requirements & Constraints



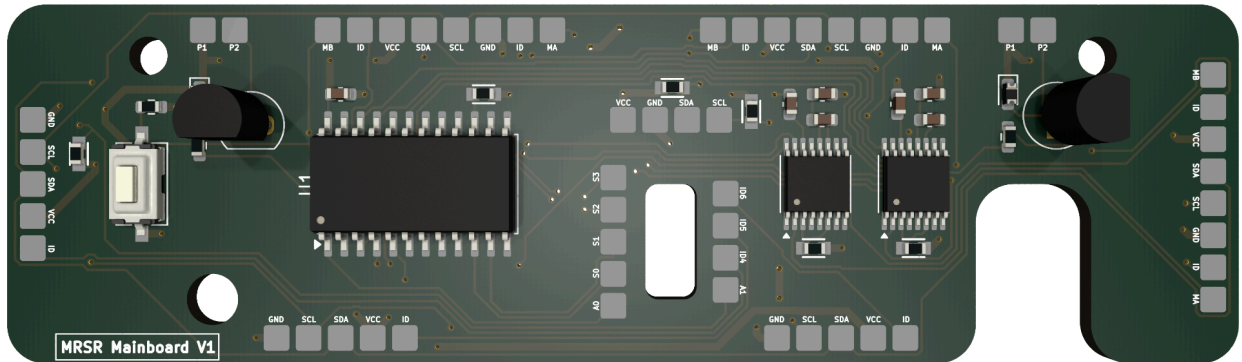
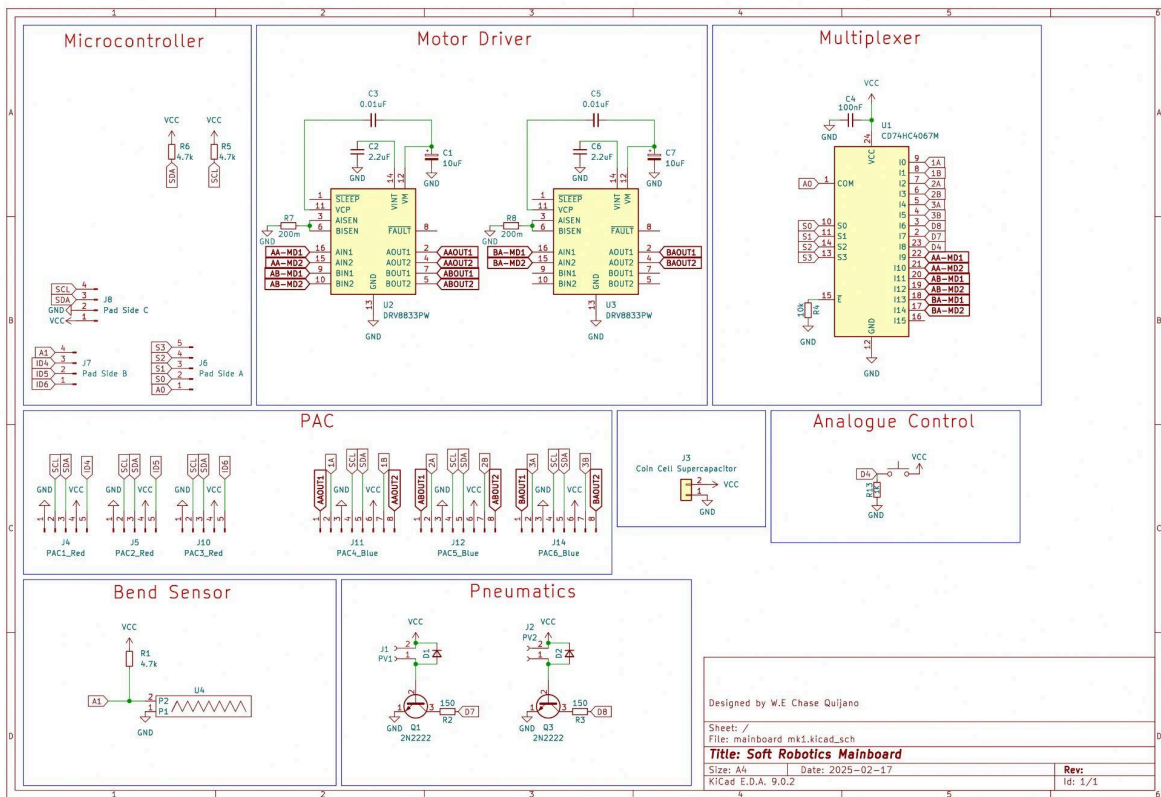
- The modular PCB footprint must fit inside each soft robotics module
- Flexible PCB capable of maintaining $\sim 1^\circ$ flexion for extended periods
- Support actuator control and sensor input
- Integration of a bending sensor for future positional data collection and testing
- Provide reliable power distribution and communication between modules
- Low-profile design for mechanical integration
- Manufacturable using flexible PCB processes; assembled in the lab

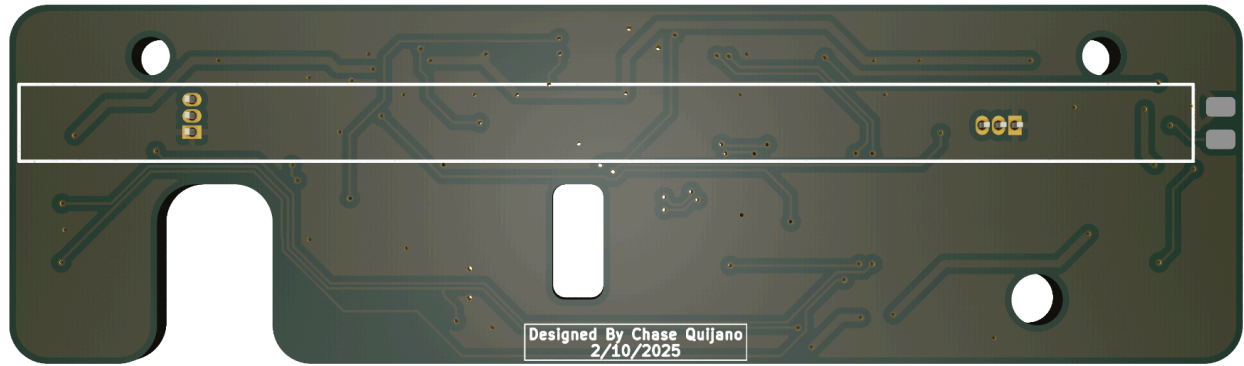
Design Approach

Component Selection

- Utilized low-profile SMD components to ensure fit within mechanical constraints
- Selected DRV8833 motor driver for reliability, availability, and current capacity matching actuator demands

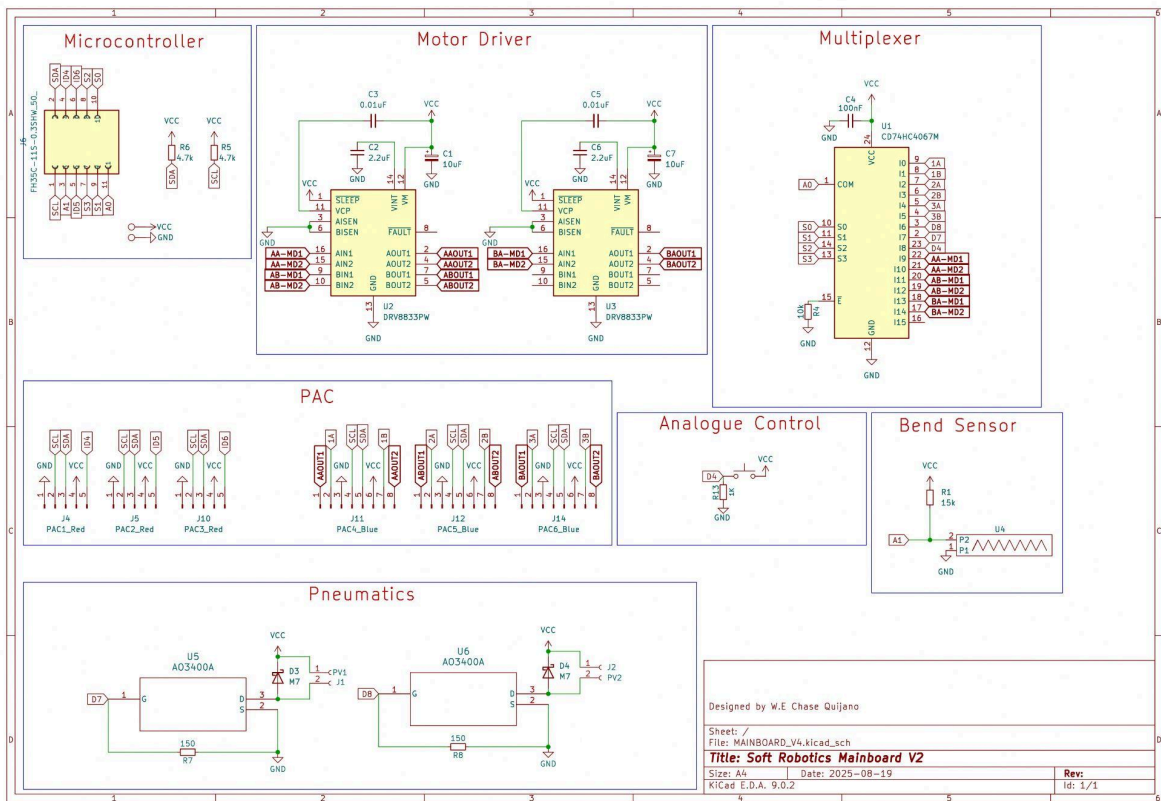
PCB Design

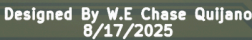




- Created schematics and two-layer PCB layouts in KiCad (to stay within research budget)
- Integrated motor drivers, power regulation, and communication interfaces

Design Iteration (Rev A → Rev B)





Sheet: /

File: Secondary Board V1.kicad_sch

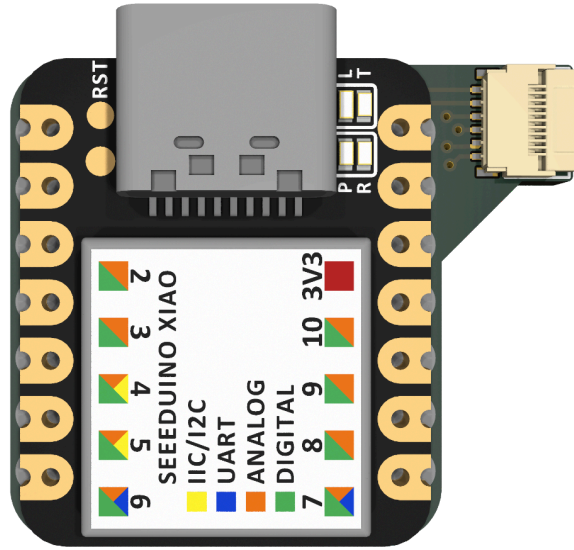
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Size: A5 Date: 8/19/2025

KiCad E.D.A. 9.0.2

Revs:

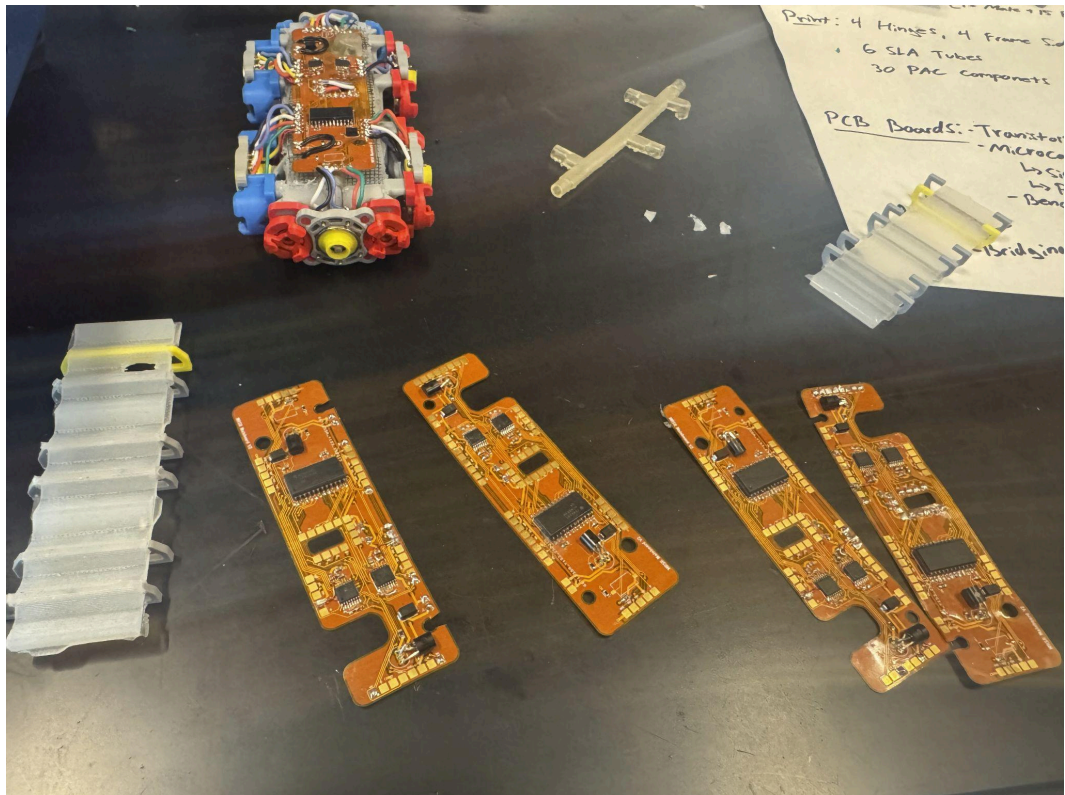
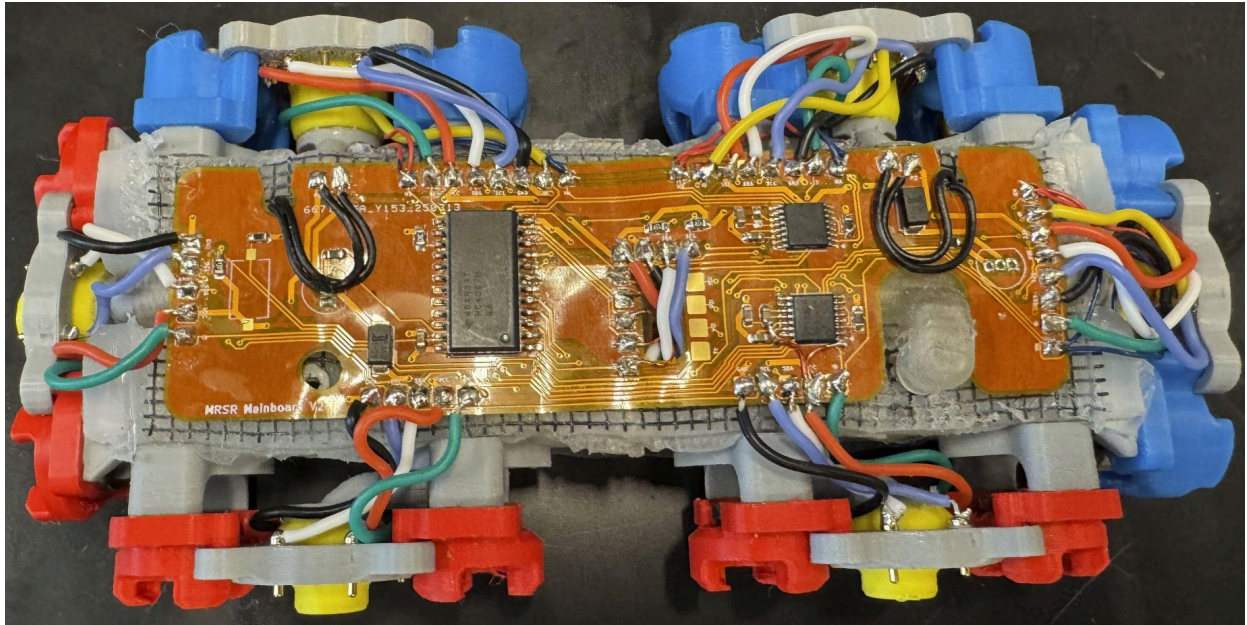
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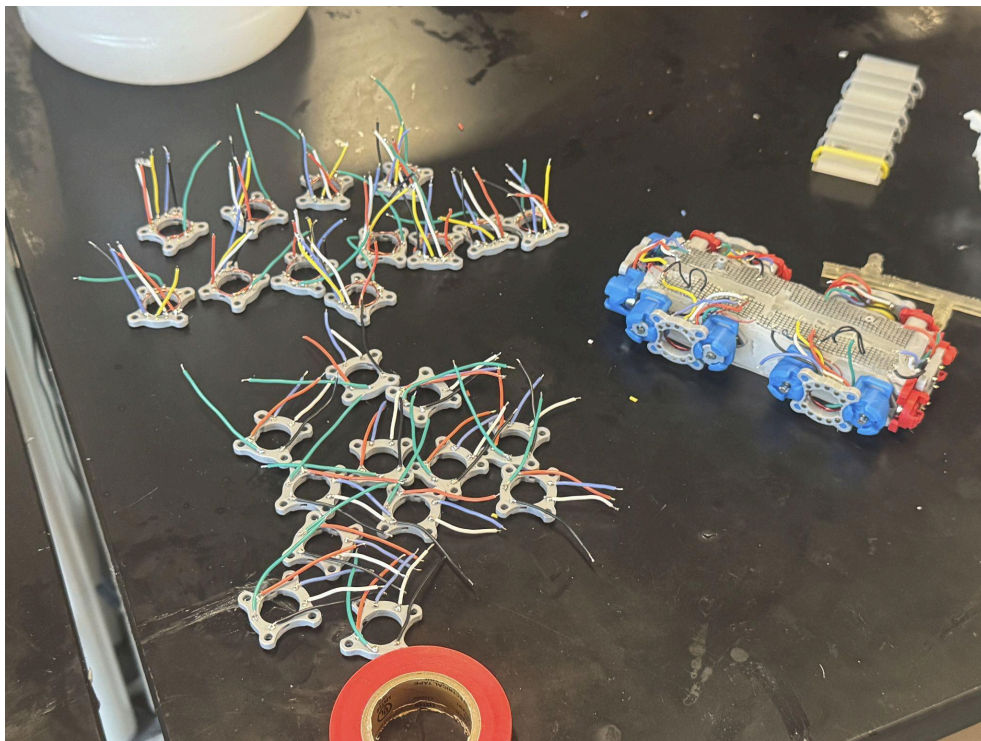
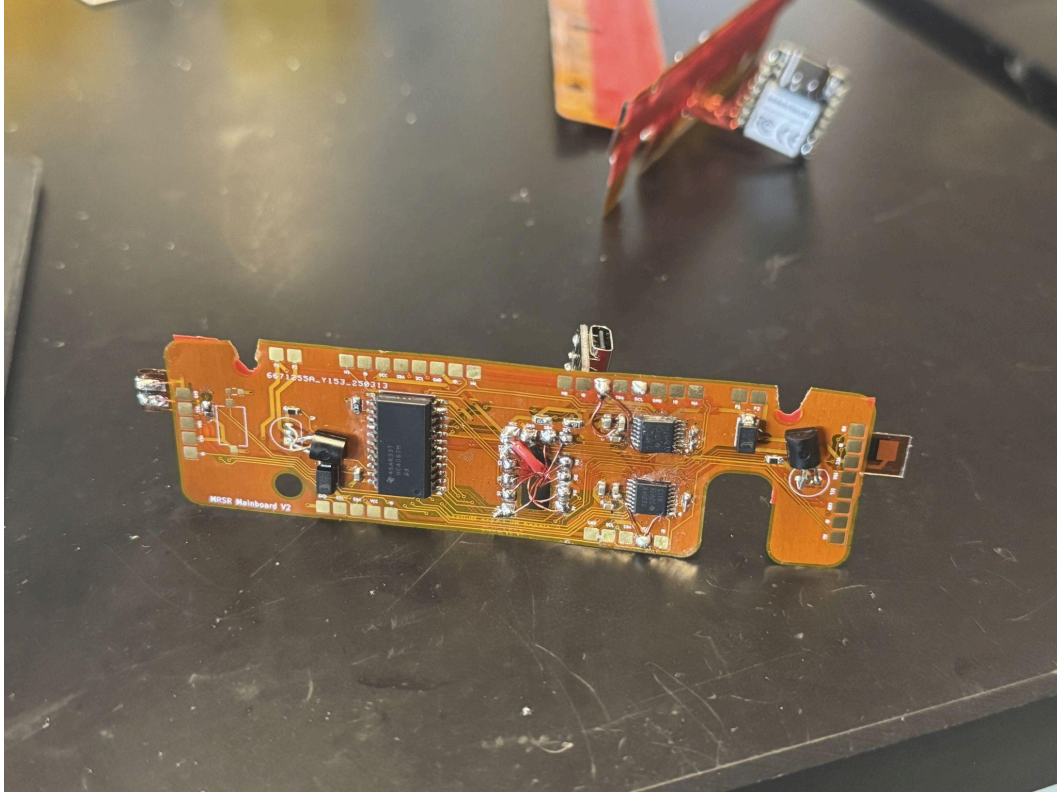


- **Rev A:** Identified 2–3 incorrect traces, requiring wire jumpers for continuity
- **Rev B:** Corrected errors, replaced through-hole transistors with SMD versions to reduce footprint and simplify assembly
- Switched from multiple discrete pads for MCU interface → compact 11-pin FPC connector (plus VCC/GND pads) for modular connection to a secondary microcontroller PCB
- Iteration improved manufacturability, modularity, and long-term reliability

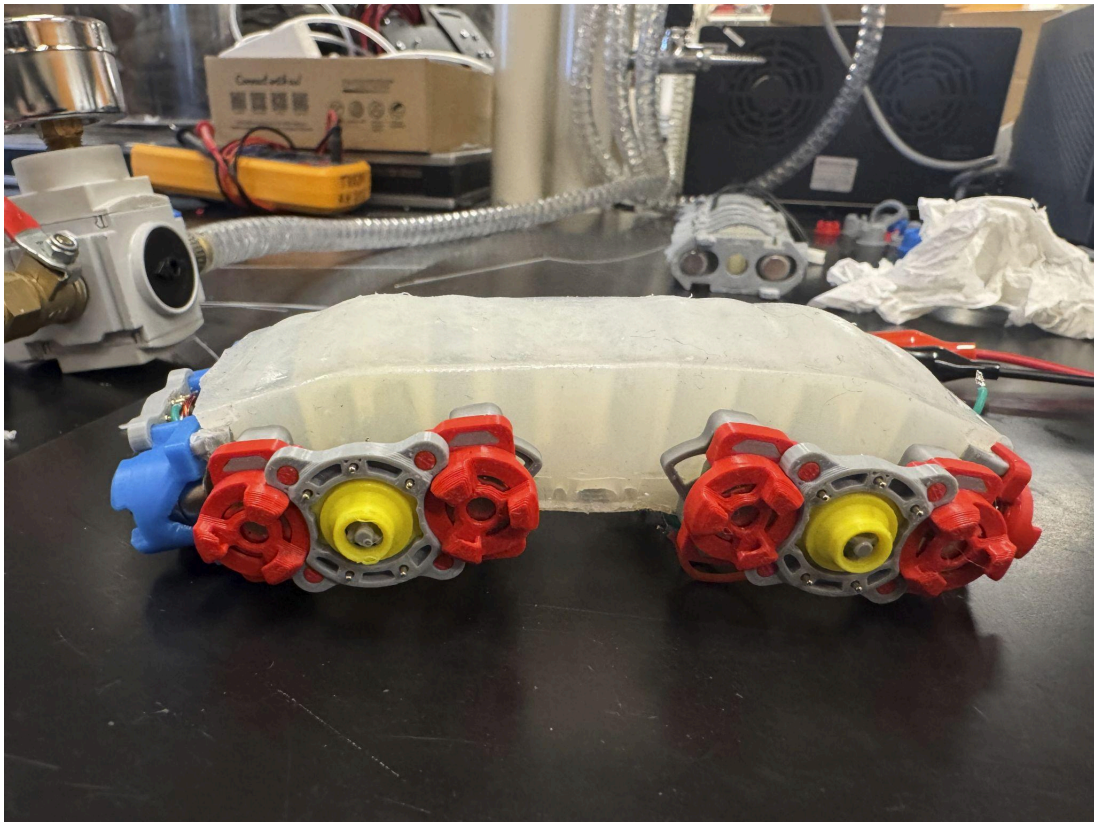
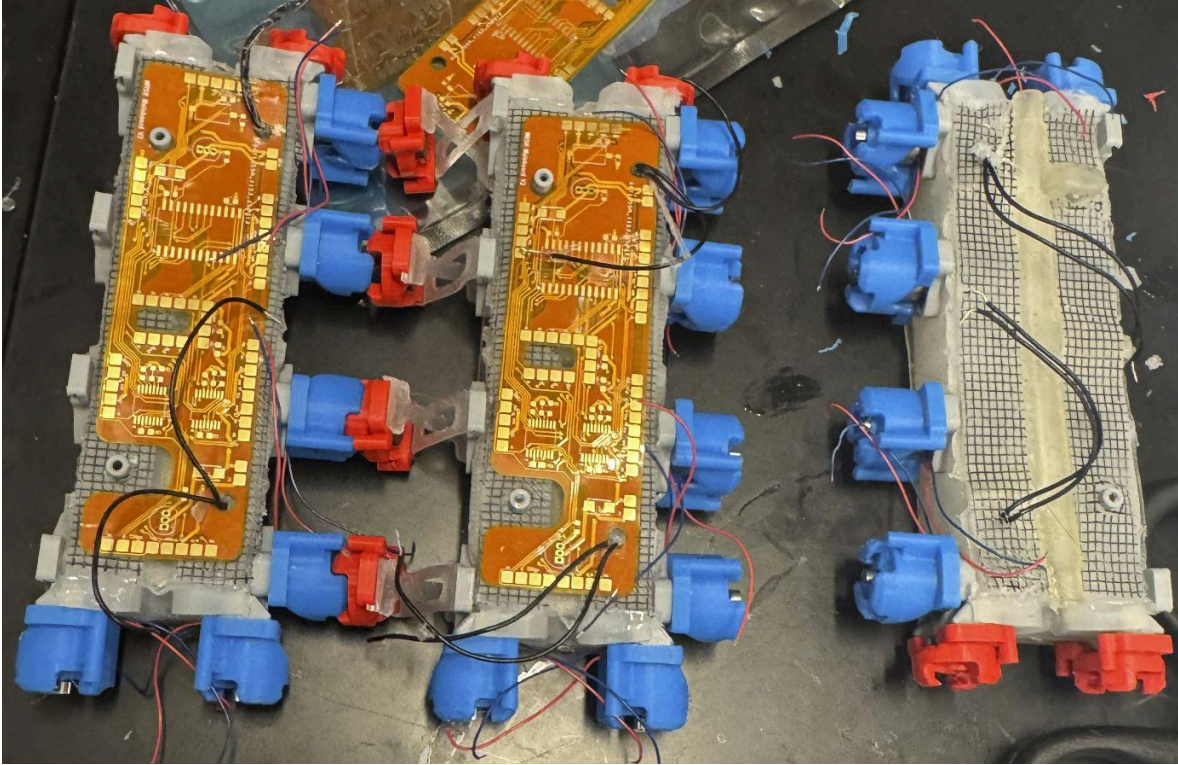
Assembly

- Hand-assembled PCBs using SMT soldering techniques (hot plate + heat gun with solder paste)
- Inspected under a microscope to ensure solder joint integrity and absence of trace bridging





PAC Connector Assembly



Module Inflation Test

Hardware Integration

- Collaborated with graduate researcher Joshua Knospler to ensure PCB form factor matched module mechanical design
- Assisted in overall module assembly and contributed to hardware design refinements

Testing & Validation

- Initial electrical validation: continuity and power integrity confirmed
 - Modules successfully powered actuators under load with stable performance
 - Full system-level validation and long-term flexion testing planned in the upcoming research phase
-

Reflection & Future Work

- Iteration from Rev A to Rev B reinforced the importance of design reviews and trace verification before fabrication
- Switching to SMD and FPC connectors improved manufacturability and modularity, lessons that will inform future designs
- Next steps:
 - Add test points for faster debugging
 - Conduct extended flexion-cycle reliability testing
 - Evaluate thermal performance under sustained load
 - Integrate fault monitoring for improved diagnostics

Research Impact

Several papers have been published by the Rowan Mechanical Engineering department, establishing the foundations of self-reconfigurable soft robotics. The PCB systems described in this case study extend that work by enabling more compact, manufacturable, and modular integration of electronics within the robotic modules. This contribution is included in a forthcoming 2025 publication, on which I am a co-author. This research is currently being evaluated for potential patenting and commercialization. While I am not directly involved in the IP process, my PCB design contributions form part of the technical foundation supporting this direction.

References

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